Introduction to Probability and Statistics

What is Statistics and Probability?

- ***Statistics*** and probability are sections of mathematics that deal with data collection and analysis.
- ***Probability*** is the study of chance and is a very fundamental subject that we apply in everyday living, while statistics is more concerned with how we handle data using different analysis techniques and collection methods.

These two subjects always go hand in hand and thus you can't study one without studying the other.

What is Statistics?

Statistics is a branch of mathematics that deals with the collection, analysis and interpretation of data.

Data can be defined as groups of information that represent the qualitative or quantitative attributes of a variable or set of variables.

Data is information collected from the population.

In layman's terms, data in statistics can be any set of information that describes a given entity.

An example of data can be the ages of the students in a given class. When you collect those ages, that becomes your data.

The population and the sample

In the language of statistics, one of the most basic concepts is ***sampling***.

Sampling is securing some elements of the population.

An *element* is a member of a population who can provide information for the population.

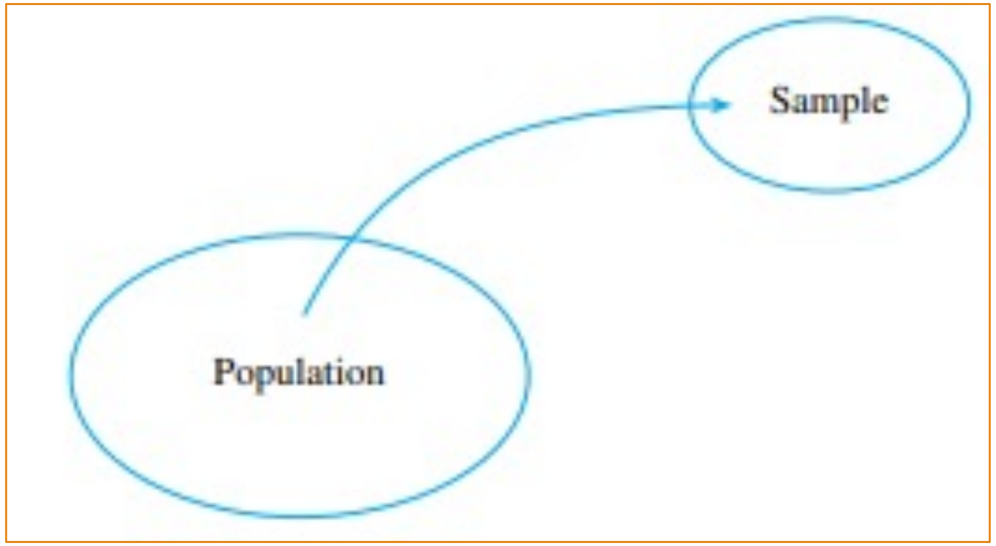
A *population* consists of the total elements about which you can make inferences based on the data gathered from a determined sample size.

In most statistical problems, a specified number of measurements or *data—a sample*—is drawn from a much larger body of measurements, called the population.

The population and the sample

In ***statistics***, a ***sample*** refers to a set of observations drawn from a population.

Often, it is necessary to use ***samples*** for research, because it is impractical to study the whole population.



Descriptive and Inferential Statistics

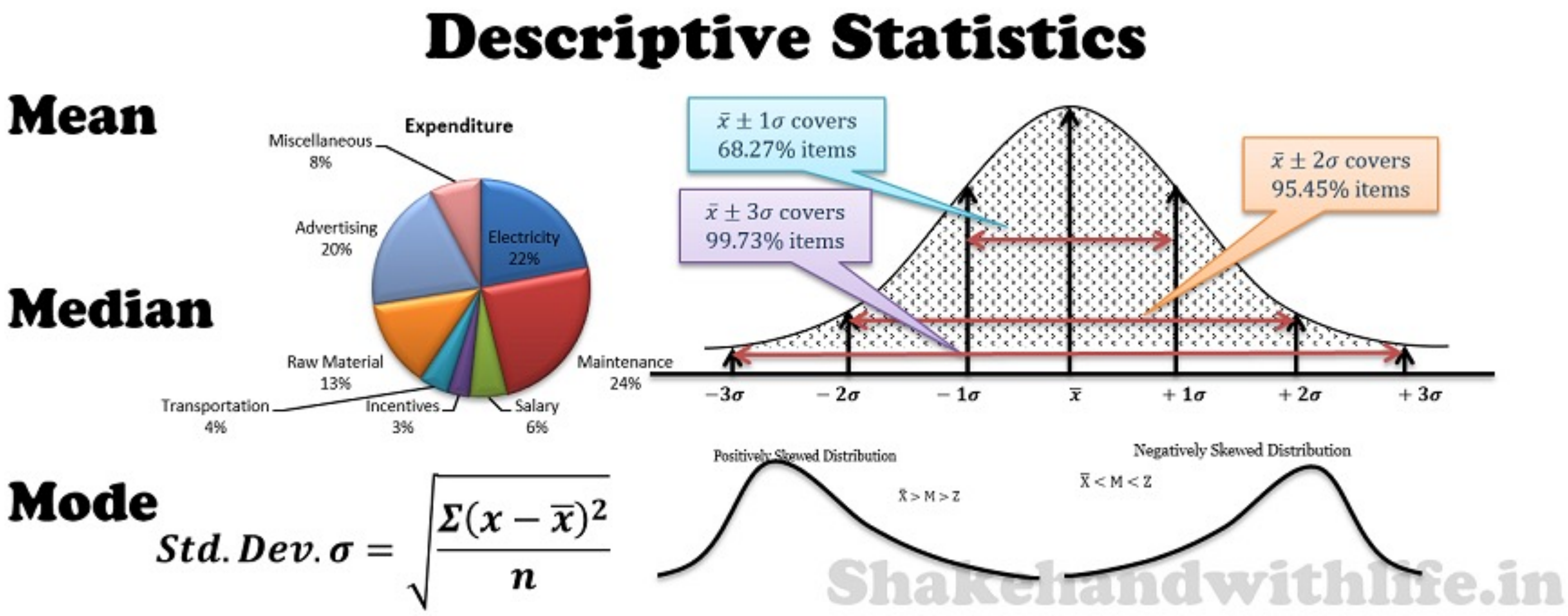
The branch of statistics that presents techniques for describing sets of measurements is called *descriptive statistics*.

You have seen descriptive statistics in many forms: bar charts, pie charts, and line charts presented by a political candidate; numerical tables in the newspaper; or the average rainfall amounts reported by the local television weather forecaster.

Formal Definition: ***Descriptive statistics*** consists of procedures used to summarize and describe the important characteristics of a set of measurements.

Descriptive Statistics

Another Definition: *Descriptive statistics* is the term given to the analysis of data that helps describe, show or summarize data in a meaningful way such that, for example, patterns might emerge from the data.



Inferential Statistics

Inferential statistics: uses statistics to make predictions.

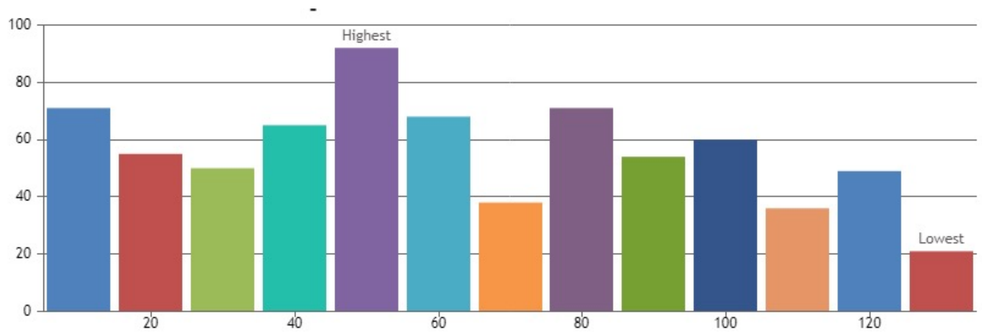
Descriptive statistics describes data (for example, a chart or graph) and **inferential statistics** allows you to make predictions ("inferences") from that data.

With inferential statistics, you take data from samples and make generalizations about a population.

For example, you might stand in a mall and ask a sample of 100 people if they like shopping at Sears.

You could make a bar chart of yes or no answers (that would be descriptive statistics) or you could use your research (and inferential statistics) to reason that around 75-80% of the population (**all** shoppers in **all malls**) like shopping at Sears.

Describing Data with Graphs



CHAPTER 1

General Objectives

In this chapter, you will learn:

A. What a *variable* is, how to classify variables into several types, and how measurements or data are generated.

B. Learn how to use graphs to describe data sets.

Topics

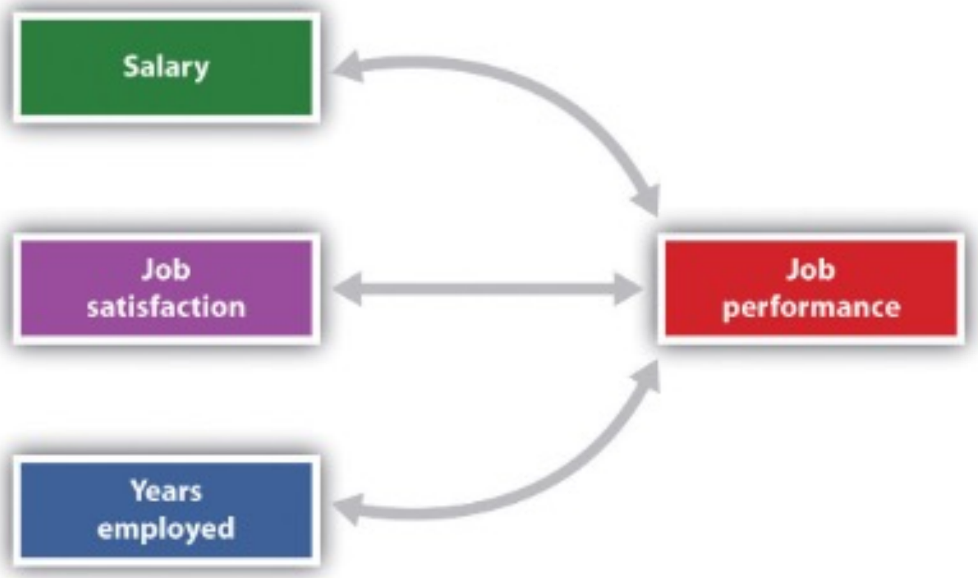
Data distributions and their shapes

- Dotplots
- Pie charts, bar charts, line charts
- Qualitative and quantitative variables—discrete and continuous
- Relative frequency histograms
- Stem and leaf plots
- Univariate and bivariate data
- Variables, experimental units, samples and populations, data

VARIABLES AND DATA

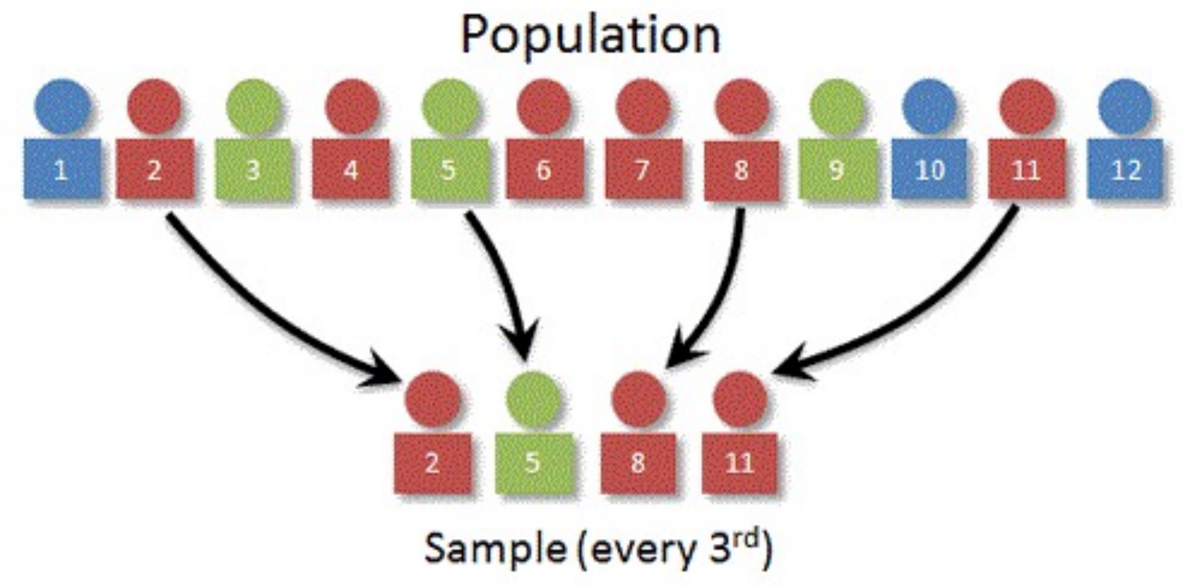
Definition: A **variable** is a characteristic that changes or varies over time and/or for different individuals or objects under consideration.

For example, body temperature is a variable that changes over time within a single individual; it also varies from person to person. Religious affiliation, ethnic origin, income, height, age, and number of offspring are all variables—characteristics that vary depending on the individual chosen.



VARIABLES AND DATA

- An *experimental unit* or an *element of the sample* as the object on which a measurement is taken.
- Equivalently, we could define an *experimental unit* as the object on which a variable is measured.



EXAMPLE 1.1

- A set of five students is selected from all undergraduates at a large university, and measurements are entered into a spreadsheet as shown in Figure 1.1.
- Identify the various elements involved in generating this set of measurements.

Worksheet 1 ***

↓	C1	C2	C3-T	C4-T	C5-T	C6
	Student	GPA	Gender	Year	Major	Number of Units
1	1	2.0	F	Fr	Psychology	16
2	2	2.3	F	So	Mathematics	15
3	3	2.9	M	So	English	17
4	4	2.7	M	Fr	English	15
5	5	2.6	F	Jr	Business	14

Figure 1.1 Measurement on 5 undergraduate students

Questions:

1. Looking at the data, what is/are the experimental unit?

2. What are the variables measured?

3. Describe each variables identified.

4. What is the result of the measurement taken in Student no. 2?

Worksheet 1 ***

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5	5	2.6	F	Jr	Business	14

Solution:

1. Looking at the data, what is/are the experimental unit?

- The *experimental unit* on which the variables are measured is a particular undergraduate student on the campus, identified in column C1.

2. What are the variables measured?

- Five variables are measured for each student: *grade point average (GPA), gender*, *year in college*, *major*, and *current number of units enrolled*.

Solution:

3. Describe each variables identified.

- Each of these characteristics varies from student to student. If we consider the GPAs of all students at this university to be the population of interest, the five GPAs in column C2 represent a *sample* from this population. If the GPA of each undergraduate student at the university had been measured, we would have generated the entire *population* of measurements for this variable.

The second variable measured on the students is gender, in column C3-T. This variable can take only one of two values—male (M) or female (F). It is not a numerically valued variable and hence is somewhat different from GPA. The population, if it could be enumerated, would consist of a set of Ms and Fs, one for each student at the university.

Similarly, the third and fourth variables, year and major, generate nonnumerical data. Year has four categories (Fr, So, Jr, Sr), and major has one category for each undergraduate major on campus. The last variable, current number of units enrolled, is numerically valued, generating a set of numbers rather than a set of qualities or characteristics.

Solution:

4. What is the result of the measurement taken in Student no. 2?

- The measurement taken on student 2 produces this observation:

(2.3, F, So, Mathematics, 15)

- **Univariate data** result when a single variable is measured on a single experimental unit.
- **Bivariate data** result when two variables are measured on a single experimental unit.
- **Multivariate data** result when more than two variables are measured.

Example:

If you measure the body temperatures of 148 people, the resulting data are *univariate.*

In Example 1.1, five variables were measured on each student, resulting in *multivariate* data.

Types of variables

Variables can be classified into one of two categories: ***qualitative*** or ***quantitative.***

1. ***Qualitative variables*** measure a quality or characteristic on each experimental unit.
2. ***Quantitative variables*** measure a numerical quantity or amount on each experimental unit.

Qualitative Variables

Qualitative variables produce data that can be categorized according to similarities or differences in kind; hence, they are often called ***categorical data*****.**

The variables gender, year, and major in Example 1.1 are qualitative variables that produce categorical data. Here are some other examples:

- Political affiliation: PDP Laban, Liberal, Independent
- Taste ranking: excellent, good, fair, poor
- Color of an M&M'S® candy: brown, yellow, red, orange, green, blue

Quantitative Variables

Quantitative variables, often represented by the letter *x,* produce numerical data, such as those listed here:

- *x* = Prime interest rate
- *x* = Number of passengers on a flight from Los Angeles to New York City
- *x* = Weight of a package ready to be shipped
- *x* = Volume of orange juice in a glass

Notice that there is a difference in the types of numerical values that these quantitative variables can assume.

To describe this difference, we define two types of quantitative variables: ***discrete*** and ***continuous***.

Definition: Discrete and Continuous

A ***discrete variable*** can assume only a finite or countable number of values.

A ***continuous variable*** can assume the infinitely many values corresponding to the points on a line interval.

The name *discrete* relates to the discrete gaps between the possible values that the variable can assume.

Variables such as number of family members, number of new car sales, and number of defective tires returned for replacement are all examples of discrete variables.

On the other hand, variables such as height, weight, time, distance, and volume are *continuous* because they can assume values at any point along a line interval.

Other definition of Discrete and Continuous

A ***discrete variable*** is a variable whose value is obtained by counting.

Examples: number of students present

number of red marbles in a jar

number of heads when flipping three coins

students' grade level

Other definition of Discrete and Continuous

A ***continuous variable*** is a variable whose value is obtained by measuring.

Examples: height of students in class

weight of students in class

time it takes to get to school

distance traveled between classes

Example 1.2

Identify each of the following variables as qualitative or quantitative:

1. The most frequent use of your microwave oven (reheating, defrosting, warming, other)

2. The number of consumers who refuse to answer a telephone survey

3. The door chosen by a mouse in a maze experiment (A, B, or C)

4. The winning time for a horse running in the Kentucky Derby

5. The number of children in a fifth-grade class who are reading at or above grade level

Answers:

- *Variables 1 and 3 are both qualitative because only a quality or characteristic is measured for each individual*. The categories for these two variables are shown in parentheses.

- *The other three variables are quantitative.*

- Variable 2, the number of consumers, is a *discrete* variable that can take on any of the values x 0, 1, 2, . . . , with a maximum value depending on the number of consumers called.

- Similarly, variable 5, the number of children reading at or above grade level, can take on any of the values x 0, 1, 2, . . . , with a maximum value depending on the number of children in the class.

- Variable 4, the winning time for a Kentucky Derby horse, is the only *continuous* variable in the list. The winning time, if it could be measured with sufficient accuracy, could be 121 seconds, 121.5 seconds, 121.25 seconds, or any values between any two times we have listed.

Why should you be concerned about different kinds of variables and the data that they generate?

- The reason is that the methods used to describe data sets depend on the type of data you have collected.
- For each set of data that you collect, the key will be to determine what type of data you have and how you can present them most clearly and understandably to your audience!

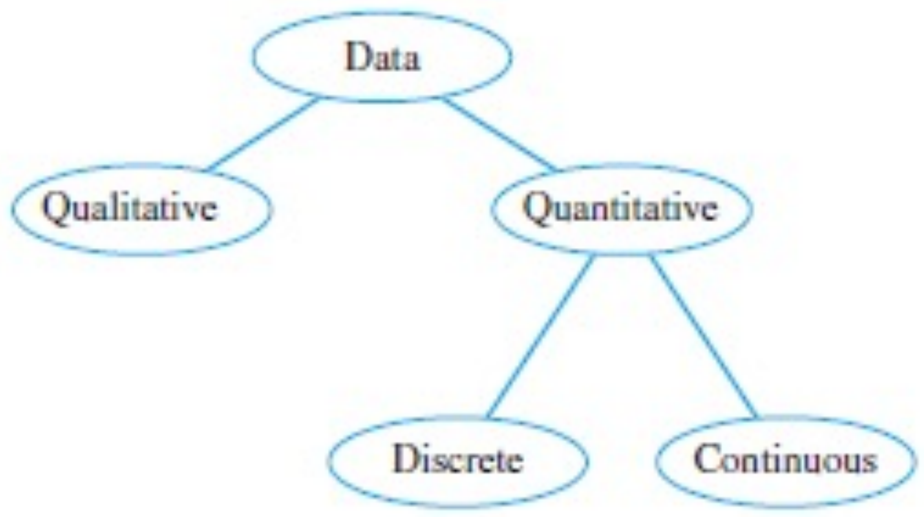


Figure 1.2: Types of data

GRAPHS FOR CATEGORICAL DATA

Graphs for Categorical data

After the data have been collected, they can be consolidated and summarized to show the following information:

- What values of the variable have been measured
- How often each value has occurred

For this purpose, you can construct a *statistical table* that can be used to display the data graphically as a data distribution. The type of graph you choose depends on the type of variable you have measured.

Graphs for Categorical data

When the variable of interest is *qualitative,* the statistical table is a list of the categories being considered along with a measure of how often each value occurred.

You can measure “how often” in three different ways:

- The **frequency,** or number of measurements in each category
- The **relative frequency,** or proportion of measurements in each category
- The **percentage** of measurements in each category

Graphs for Categorical data

For example, if you let *n* be the total number of measurements in the set, you can find the relative frequency and percentage using these relationships:

$$\text{Relative frequency} = \frac{\text{Frequency}}{n}$$

$$\text{Percent} = 100 \times \text{Relative frequency}$$

You will find that the sum of the frequencies is always *n,* the sum of the relative frequencies is 1, and the sum of the percentages is 100%.

The categories for a qualitative variable should be chosen so that

• a measurement will belong to one and only one category

• each measurement has a category to which it can be assigned

Example

For example, if you categorize meat products according to the *type of meat used*, you might use these *categories: beef, chicken, seafood, pork, turkey, other*.

To categorize *ranks of college faculty*, you might use these *categories: professor, associate professor, assistant professor, instructor, lecturer, other.*

The "other" category is included in both cases to allow for the possibility that a measurement cannot be assigned to one of the earlier categories.

Use of Charts

Once the measurements have been categorized and summarized in a *statistical table, you can use either a pie chart or a bar chart to display the distribution of the data*.

A ***pie chart*** is the familiar circular graph that shows how the measurements are distributed among the categories.

A ***bar chart*** shows the same distribution of measurements in categories, with the height of the bar measuring how often a particular category was observed.

EXAMPLE 1.3

In a survey concerning public education, 400 school administrators were asked to rate the quality of education in the United States.

Their responses are summarized in Table 1.1. Construct a pie chart and a bar chart for this set of data.

TABLE 1.1 U.S. Education Rating by 400 Educators

Rating	Frequency
A	35
B	260
C	93
D	12
Total	400

Solution

To construct a pie chart, assign one sector of a circle to each category.

The angle of each sector should be proportional to the proportion of measurements (or *relative frequency*) in that category.

Since a circle contains 360°, you can use this equation to find the angle:

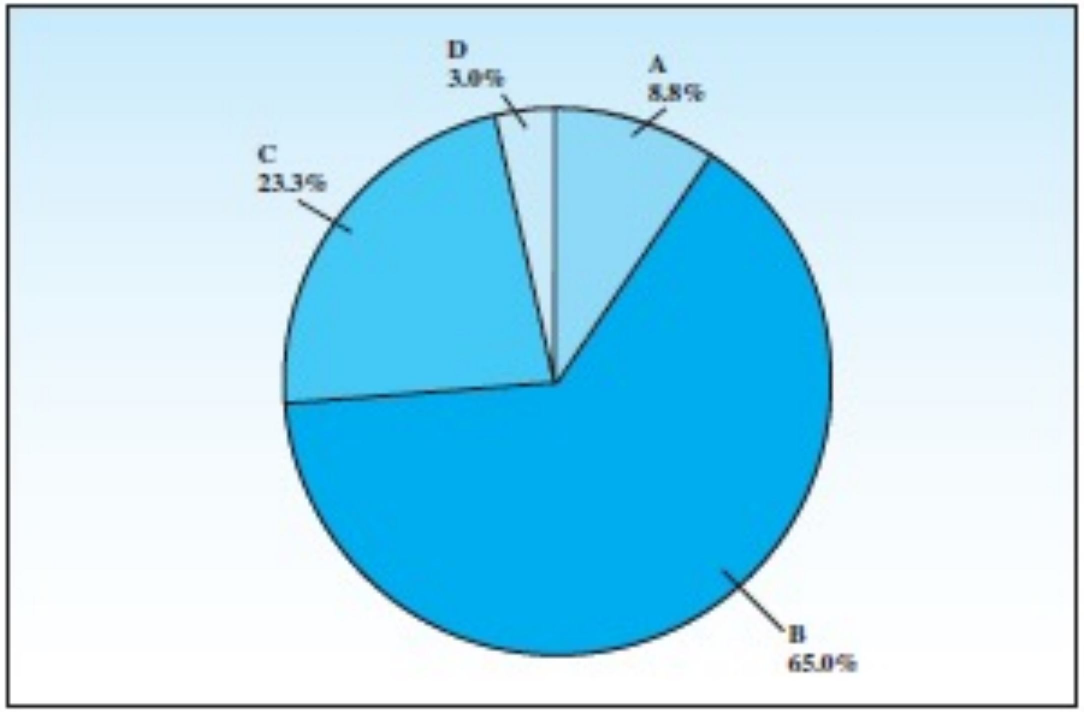
$$\text{Angle} = \text{Relative frequency} \times 360^\circ$$

Calculations for the Pie Chart in Example 1.3

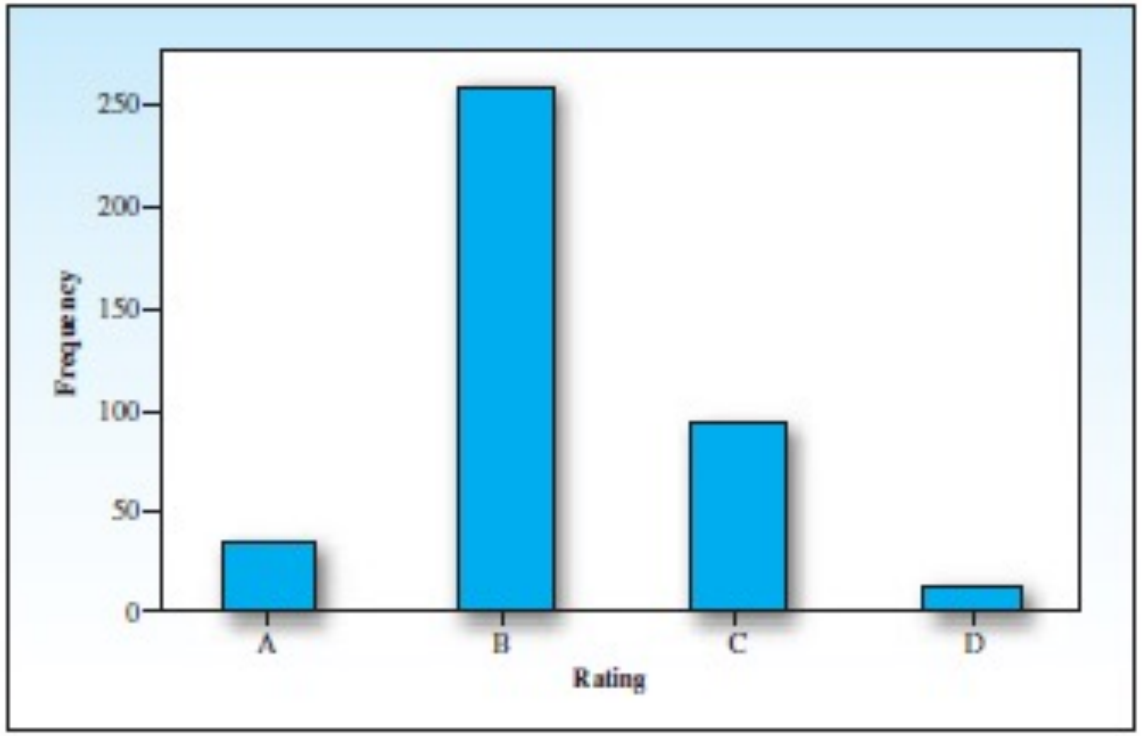
Rating	Frequency	Relative Frequency	Percent	Angle
A	35	$35/400 = .09$	9%	$.09 \times 360 = 32.4^\circ$
B	260	$260/400 = .65$	65%	234.0°
C	93	$93/400 = .23$	23%	82.8°
D	12	$12/400 = .03$	3%	10.8°
Total	400	1.00	100%	360°

The visual impact of these two graphs is somewhat different. The pie chart is used to display the relationship of the parts to the whole; the bar chart is used to emphasize the actual quantity or frequency for each category.

Pie Chart for Example 1.3



Bar Chart for Example 1.3



GRAPHS FOR QUANTITATIVE DATA

Graphs for Quantitative Data

Quantitative variables measure an amount or quantity on each experimental unit.

If the variable can take only a finite or countable number of values, it is a *discrete variable*.

A variable that can assume an infinite number of values corresponding to points on a line interval is called *continuous*.

Pie Charts and Bar Charts

Sometimes information is collected for a quantitative variable measured on different segments of the population, or for different categories of classification.

For example, you might measure the average incomes for people of different age groups, different genders, or living in different geographic areas of the country.

In such cases, *you can use pie charts or bar charts to describe the data, using the amount measured in each category rather than the frequency of occurrence of each category*.

The pie chart displays how the total quantity is distributed among the categories, and the bar chart uses the height of the bar to display the amount in a particular category.

Example 1.5

The amount of money expended in fiscal year 2005 by the U.S. Department of Defense in various categories is shown in Table 1.5.

Construct both a pie chart and a bar chart to describe the data. Compare the two forms of presentation.

Expenses by Category

Category	Amount (in billions)
Military personnel	$127.5
Operation and maintenance	188.1
Procurement	82.3
Research and development	65.7
Military construction	5.3
Other	5.5
Total	$474.4

Source: The World Almanac and Book of Facts 2007

Solution

Two variables are being measured:

- the category of expenditure (qualitative) and
- the amount of the expenditure (quantitative).

The bar chart in Figure 1.6 displays the categories on the horizontal axis and the amounts on the vertical axis.

For the pie chart in Figure 1.7, each “pie slice” represents the proportion of the total expenditures (\$474.4 billion) corresponding to its particular category.

Solution

For example, for the research and development category, the angle of the sector is

$$\frac{65.7}{474.4} \times 360° = 49.9°$$

Both graphs show that the largest amounts of money were spent on personnel and operations.

Since there is no inherent order to the categories, you are free to rearrange the bars or sectors of the graphs in any way you like. The shape of the bar chart has no bearing on its interpretation.

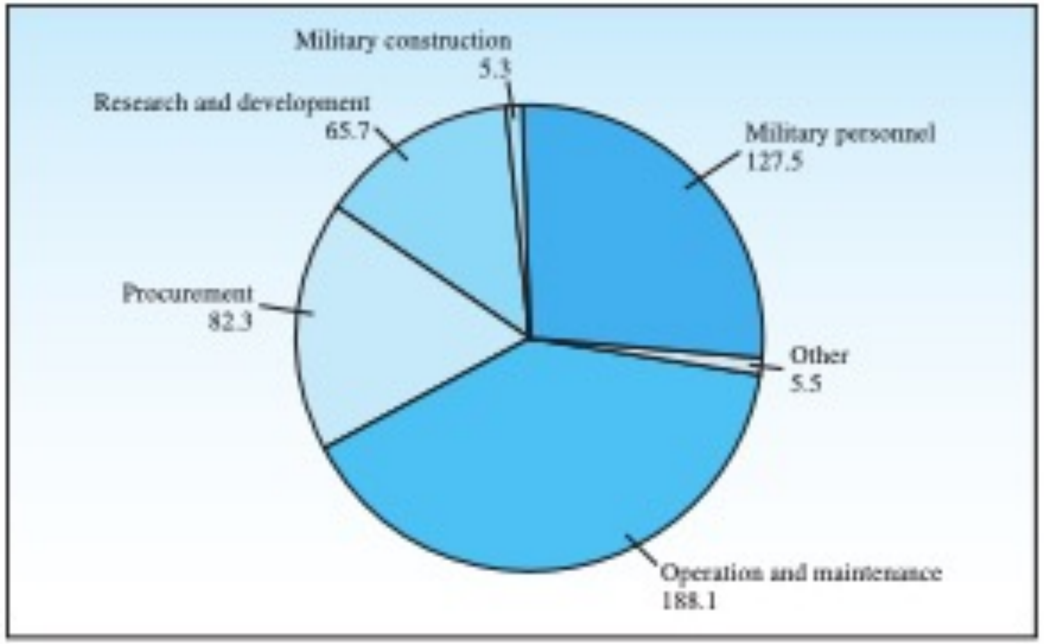


Figure 1.7 Pie Chart for Example 1.5

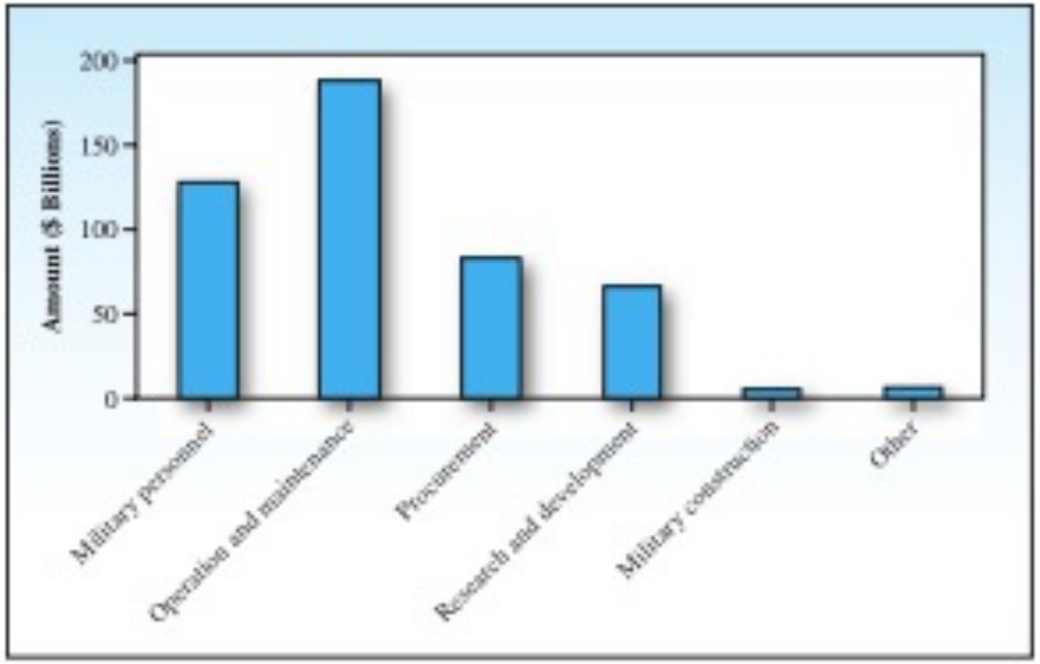


Figure 1.6 Bar Chart for Example 1.5

Pie Chart and Bar Chart

Line Charts

When a *quantitative variable* is recorded over time at equally spaced intervals (such as daily, weekly, monthly, quarterly, or yearly), the data set forms a time series.

Time series data are most effectively presented on a line chart with time as the horizontal axis.

The idea is to try to discern a *pattern or trend* that will likely continue into the future, and then to use that pattern to make accurate predictions for the immediate future.

Example 1.6

In the year 2025, the oldest “baby boomers” (born in 1946) will be 79 years old, and the oldest “Gen-Xers” (born in 1965) will be two years from Social Security eligibility.

How will this affect the consumer trends in the next 15 years? Will there be sufficient funds for “baby boomers” to collect Social Security benefits? The United States Bureau of the Census gives projections for the portion of the U.S. population that will be 85 and over in the coming years, as shown below.

Construct a line chart to illustrate the data. What is the effect of stretching and shrinking the vertical axis on the line chart?

Population Growth Projections

Year	2010	2020	2030	2040	2050
85 and over (millions)	6.1	7.3	9.6	15.4	20.9

Solution

The quantitative variable “85 and over” is measured over five time intervals, creating a time series that you can graph with a line chart.

The time intervals are marked on the horizontal axis and the projections on the vertical axis.

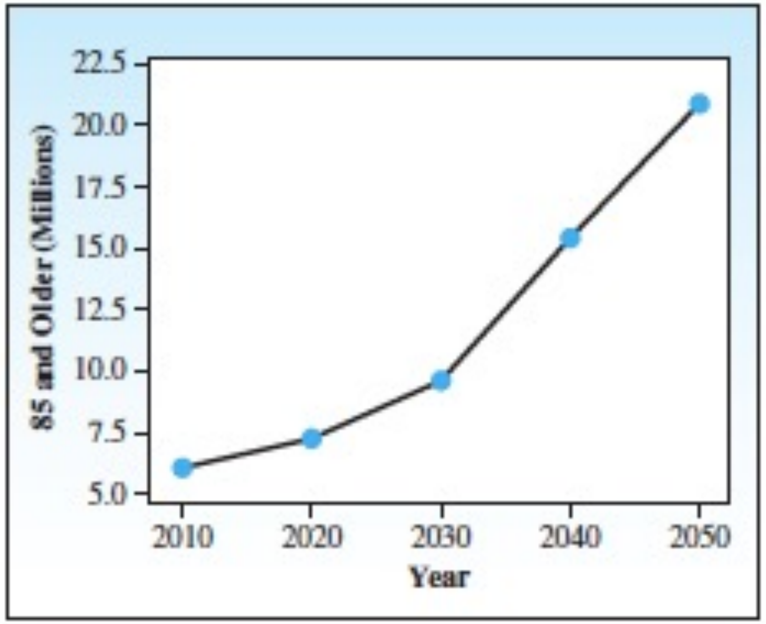
The data points are then connected by line segments to form the line charts in Figure 1.8.

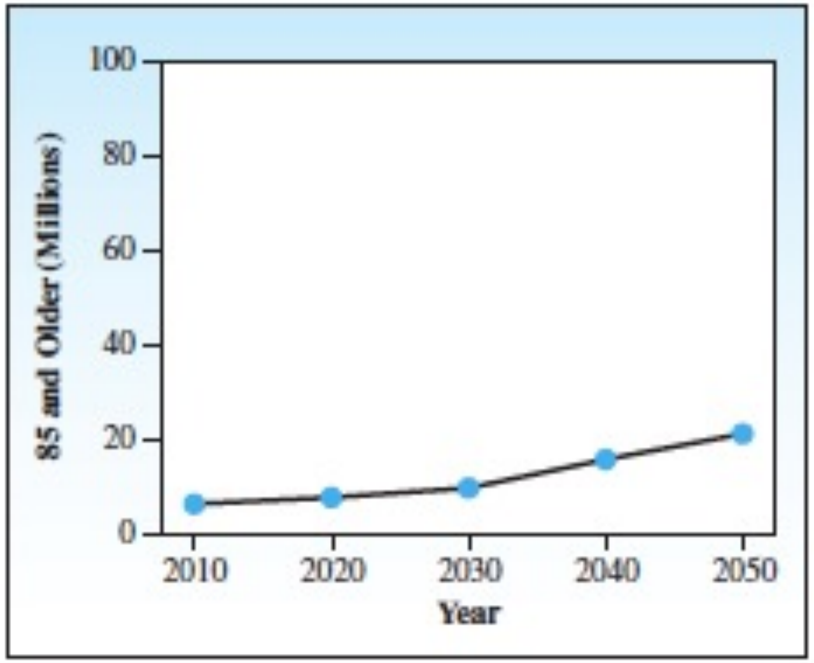
Solution

Shrinking the scale on the vertical axis causes large changes to appear small, and vice versa.

To avoid misleading conclusions, you must look carefully at the scales of the vertical and horizontal axes.

However, from both graphs you get a clear picture of the steadily increasing number of those 85 and older in the early years of the new millennium.





Dotplots

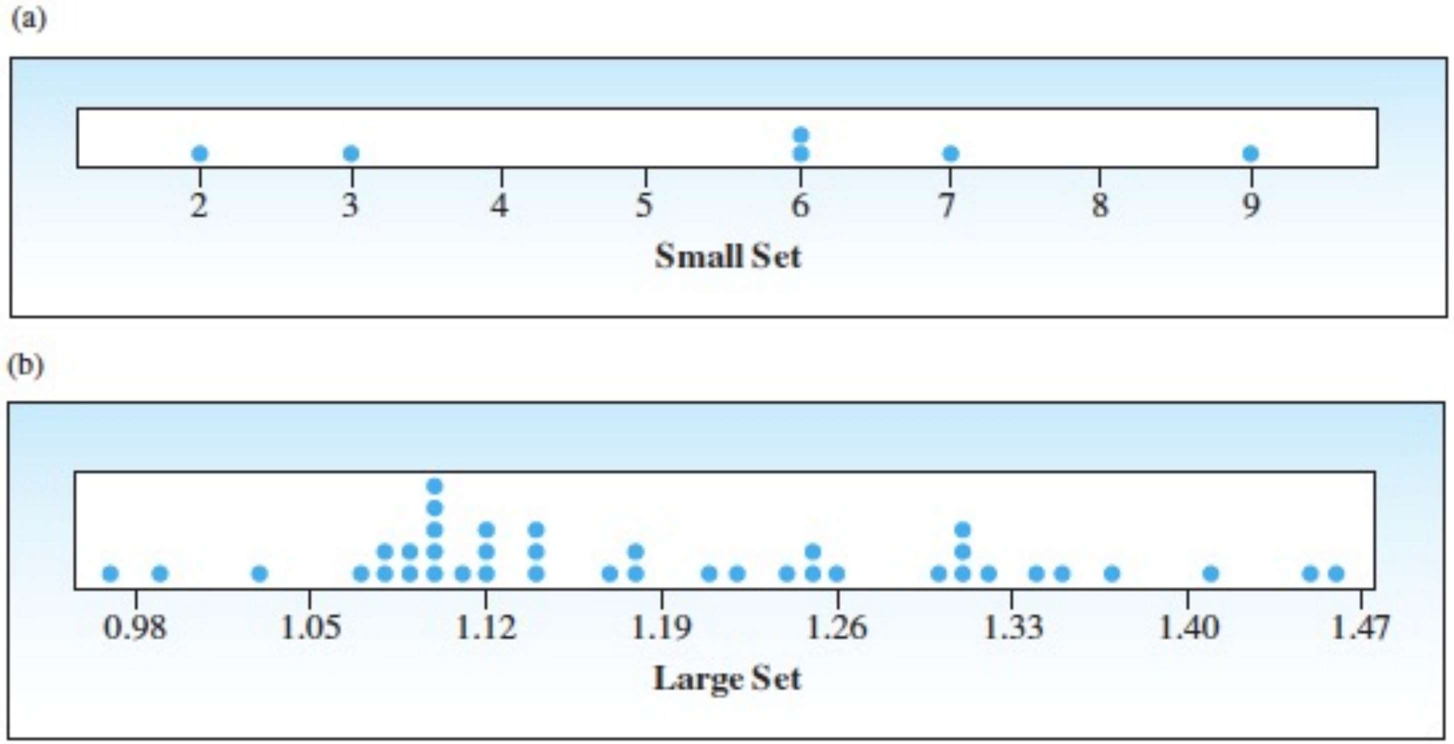
Many sets of quantitative data consist of numbers that cannot easily be separated into categories or intervals of time.

You need a different way to graph this type of data!

The simplest graph for quantitative data is the *dotplot*.

Example

For a small set of measurements— for example, the set 2, 6, 9, 3, 7, 6—you can simply plot the measurements as points on a horizontal axis.



Dotplots for small and large data sets

Stem and Leaf Plots

Another simple way to display the distribution of a quantitative data set is the ***stem and leaf plot***.

This plot presents a graphical display of the data using the actual numerical values of each data point.

EXAMPLE 1.7

Table 1.7 lists the prices (in dollars) of 19 different brands of walking shoes. Construct a stem and leaf plot to display the distribution of the data.

Prices of Walking Shoes

90	70	70	70	75	70
65	68	60	74	70	95
75	70	68	65	40	65
70					

TABLE 1.7

Solution

To create the stem and leaf, you could divide each observation between the ones and the tens place. The number to the left is the stem; the number to the right is the leaf.

Thus, for the shoes that cost $65, the stem is 6 and the leaf is 5. The stems, ranging from 4 to 9, are listed in Figure 1.10, along with the leaves for each of the 19 measurements.

If you indicate that the leaf unit is 1, the reader will realize that the stem and leaf 6 and 8, for example, represent the number 68, recorded to the nearest dollar.

4	0
5	
6	5 8 0 8 5 5
7	0 0 0 5 0 4 0 5 0 0
8	
9	0 5

Leaf unit = 1

$\boxed{\text{Reordering}} \longrightarrow$

4	0
5	
6	0 5 5 5 8 8
7	0 0 0 0 0 0 0 4 5 5
8	
9	0 5

Figure 1.10. Stem and leaf plot for the data in Table 1.7

Activity: (Example 1.8)

The data in Table 1.8 are the weights at birth of 30 full-term babies, born at a metropolitan hospital and recorded to the nearest tenth of a pound.

Construct a stem and leaf plot to display the distribution of the data.

Birth Weights of 30 Full-Term Newborn Babies

7.2	7.8	6.8	6.2	8.2
8.0	8.2	5.6	8.6	7.1
8.2	7.7	7.5	7.2	7.7
5.8	6.8	6.8	8.5	7.5
6.1	7.9	9.4	9.0	7.8
8.5	9.0	7.7	6.7	7.7

TABLE 1.8

Solution

Steam and Leaf plot for Figure 1.8

5	8 6		5	6 8
6	1 2		6	1 2
6	8 8 8 7	$\boxed{\text{Reordering}} \rightarrow$	6	7 8 8 8
7	2 2 1		7	1 2 2
7	8 7 9 5 7 7 5 8 7		7	5 5 7 7 7 7 8 8 9
8	0 2 2 2		8	0 2 2 2
8	5 6 5		8	5 5 6
9	0 4 0	Leaf unit = .1	9	0 0 4

Interpreting Graphs with a Critical Eye

A *distribution is symmetric* if the left and right sides of the distribution, when divided at the middle value, form mirror images.

A distribution is *skewed to the right* if a greater proportion of the measurements lie to the right of the peak value.

Distributions that are *skewed right* contain a few unusually large measurements.

Interpreting Graphs with a Critical Eye

A distribution is *skewed to the left* if a greater proportion of the measurements lie to the left of the peak value.

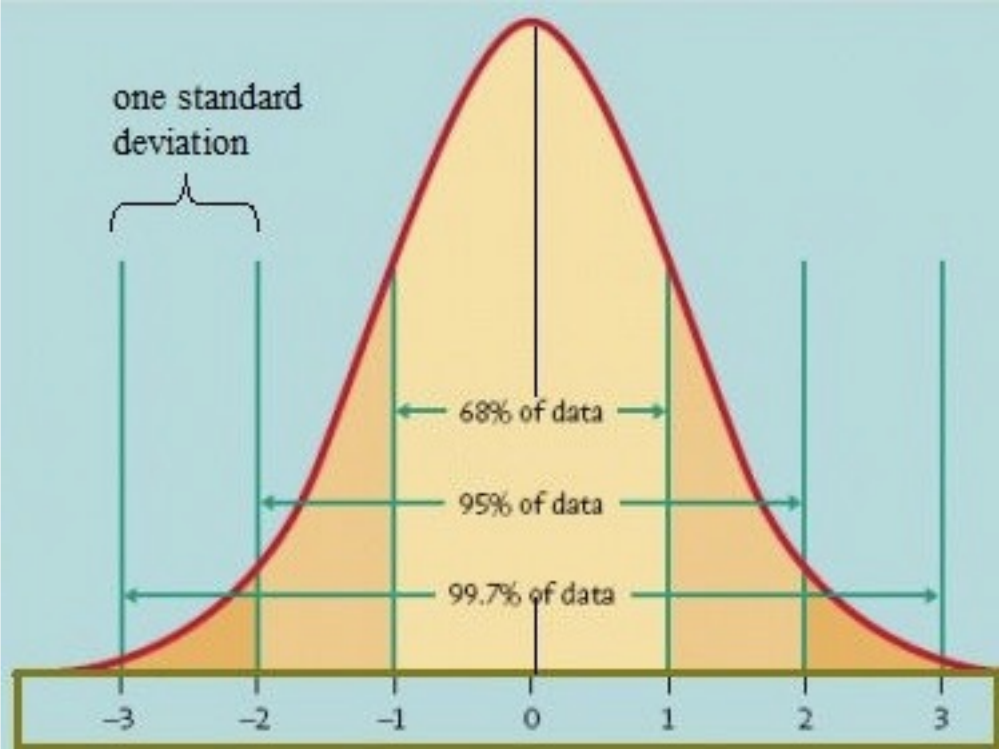
Distributions that are *skewed left* contain a few unusually small measurements.

A *distribution* is *unimodal if it has one peak*; a bimodal distribution has two peaks.

Bimodal distributions often represent a mixture of two different populations in the data set.

What type of distribution is this?

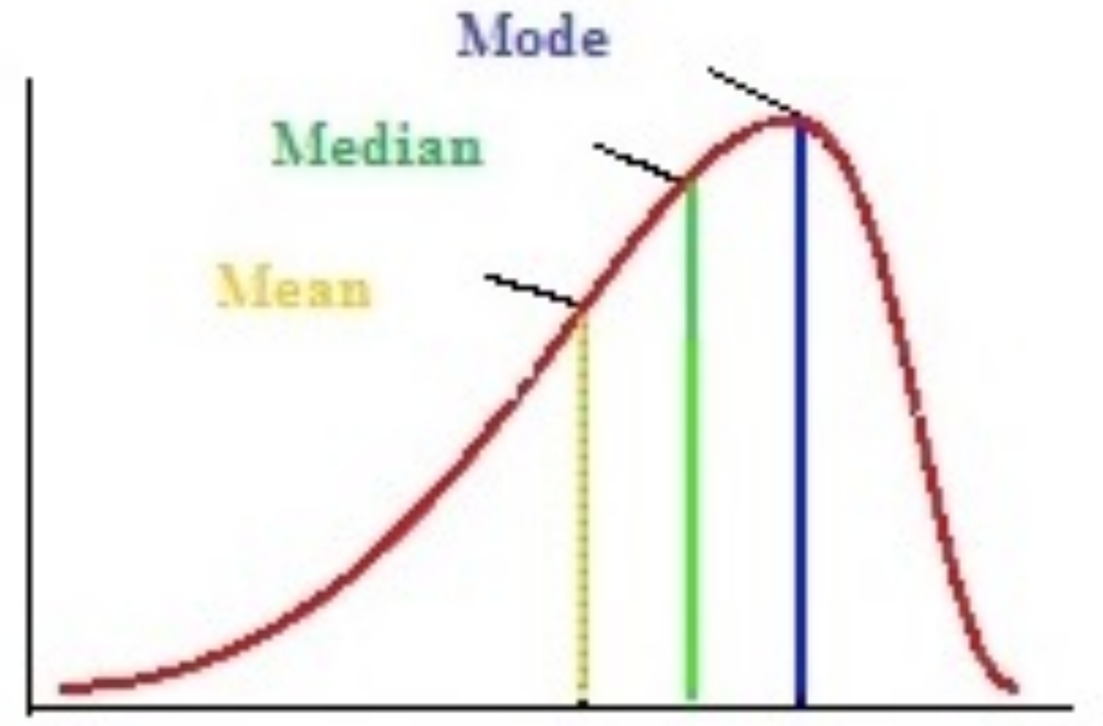
Answer: Symmetric Distribution



What type of distribution is this?

Answer: Skewed to the left

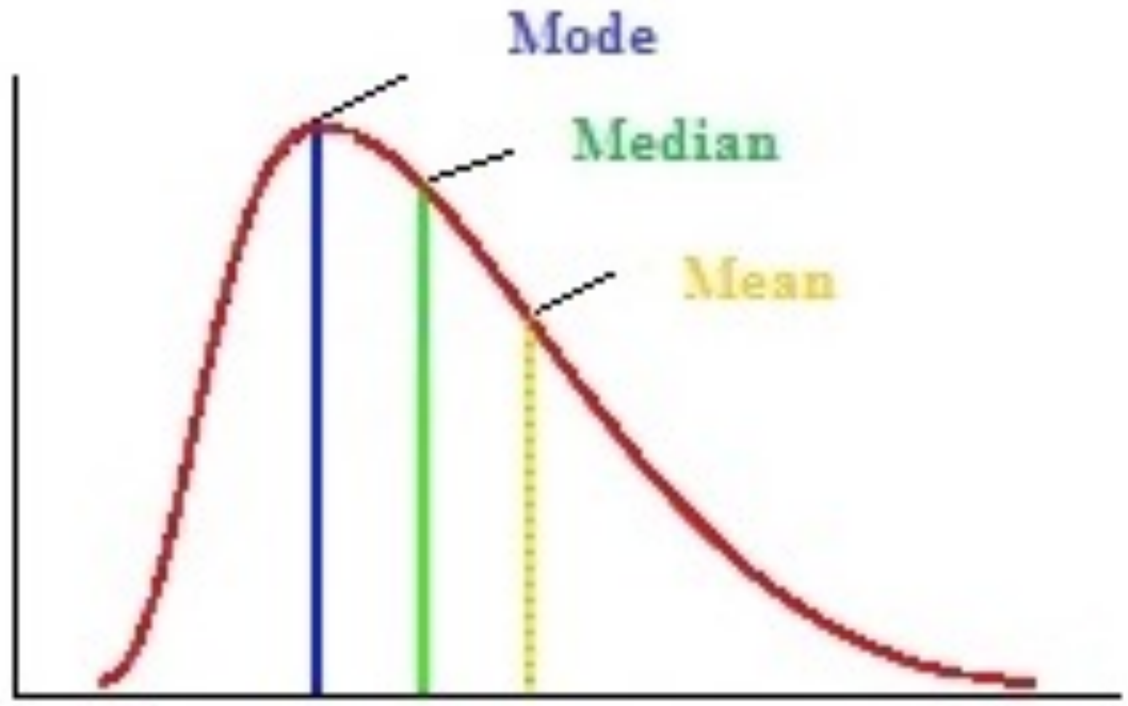
(Negatively Skewed)



What type of distribution is this?

Answer: Skewed to the right

(Positively Skewed)



Chapter Outline

I. How Data Are Generated

1. Experimental units, variables, measurements

2. Samples and populations

3. Univariate, bivariate, and multivariate data

II. Types of Variables

1. Qualitative or categorical

2. Quantitative
   - a. Discrete
   - b. Continuous

Chapter Outline

III. Graphs for Univariate Data Distributions

1. Qualitative or categorical data

a. Pie charts

b. Bar charts

2. Quantitative data

a. Pie and bar charts

b. Line charts

c. Dotplots

d. Stem and leaf plots

3**. Describing data distributions**

a. Shapes—symmetric, skewed left, skewed right, unimodal, bimodal

THANK YOU!